ABSOLUTE EXTREMA

The Extreme Value Theorem

Click a hypothesis to see what breaks without it.

IF

1. the interval is closed and bounded
   CLICK: DROP IT
2. f is continuous on the interval
   CLICK: DROP IT

$\Longrightarrow$

THEN

f achieves both an absolute maximum and an absolute minimum

The extrema occur either at local extrema within the interval or at the endpoints.